Summary

Scope. Every row of Tables A, B and C, and every section outside them (n37, n9 F6-IC, the F6-IC ladder, F4/F5/F6, n41/n53, conductor-gap, N83 holdout), plus two groups that exist only in `leaderboard.json` (the oracle-pricing cells and the linear-algebra engine rows) and the five frontier-audit statements. 55 verdict records in `verdicts.jsonl`.

This audit repairs nothing and changes no research state. It does not regenerate the leaderboard. Rows that should be demoted are listed under "Demotion flags".

verdict	rows
validated	1
validated_with_caveats	45
unvalidated	8
contradicted	1

Unvalidated: the three F6-IC ladder rows, the four F4/F5/F6 cross-family rows, and the n61 online bar. Contradicted: the frontier root-finder replay-block statement. No Table A, B or C row is below validated_with_caveats.

What held.

- Every published number in Tables A, B and C recomputes exactly from the raw records (all digits). So do the n37, F6-IC ladder, F4/F5/F6, n41/n53, conductor-gap, N83 and linear-algebra figures I could reach from raw records (`recomputed_ok`).
- Every sampled recovered scalar verifies. An independent verifier, written for this audit and sharing no code with the repository, checked [d]G = Q and [r]G = O on about 95 seeded records over GF(2^n) and GF(p), including 5 of 5 N83 points on GF(2^83). A negative control (scalar + 1) fails as it should.
- The leaderboard's own central claim, that no row is below its reference, survives everything below. Where I priced the reference differently, IC stays above rho.

What did not hold, in order of weight.

1. **Table A's reference rho is not priced like its IC arm.** The IC unit is one 1024-lane batched addition; rho runs 32 lanes and costs 9.7 to 11.8 units a step against a canonical step of 4.6 to 5.4. At the canonical step the published x reference of 5.2 to 21 becomes 9.1 to 41, so the gap to rho is understated by 1.7 to 2.0 times. Table C prices its rho at the canonical step, so A and C use different rho.
2. **The inherited-F4 and F6-IC counters do not replay on current main.** See "Known failure": on this host the F6-IC ladder sessions reproduce 24 of 72 replays at m = 13 and at m = 19, and the root-finder session reproduces 72 of 168, not 120 of 168. The failure is wider than the 48 inherited-F4 rows the board names.
3. **Section 5a was not applied.** No row carries the 0 / 0.1 / 1 band. I computed it: Table B's order holds at every weight (IC stays 4.7 to 55 times rho), and the repository's own `ecbench band` makes every n37 IC pair "unrankable (uncounted unpriced work)".
4. **Table C's reference is from a different experiment** (S20, 2026-09-26) from its IC figure (S22 isolated, 2026-09-30), re-checked at one size only.
5. **Freeze order is not demonstrable for 6 evidence groups** (squashed PRs put protocol and data in one commit).
6. **One admitted row has no locatable evidence:** the n61 compact-orbit online bar.

Method and model split

Judgement (rho fairness, pricing, classification, verdicts) was done by the Sonnet 5.5 session. Three mechanical cards ran on Haiku subagents with frozen commands: the provenance cross-check, the independent re-run of the three recompute scripts, and the seeded Table A replay. I diffed the first two against my own runs: byte-identical. The Table A replay output was reviewed row by row. Replays too large for a subagent (native ladder replays, `ecbench band`) were run by the session itself.

The replay seed was committed before any replay (`REPLAY_SEED.md`, commit 0cb3bc9e3). Runs are drawn by `sha256(seed|row|run)`; no run was dropped or swapped. One draw was wrong: the `n53:ic-k16` row of the session replay returned records of the n41 curve, because ic-k16 never verified at n = 53 (0 of 80). I removed it from the n53 row's replay count and say so in that row.

Conductor: `conductor check` reported the scope clear; `coord_start_work` returned HTTP 500 three times, so the session was not registered as a claim.

Known failure: which rows do the root-finder counters feed?

Published: 120 of 168 replays identical, 48 inherited-F4 rows differing in solver phase counts (session `ECBS1hfd5bf1885227`).

Fresh replay-all, this host (x86_64 Linux), binary built from main `cb2ec6873` (`raw/replay_all/`):

session	replays identical
`ecbench_rootfind_engine_20261012__claim-n13`	72 / 168
`f4_f5_f6_cross_family_20261011__K13`	16 / 48
`f4_f5_f6_cross_family_20261011__K19`	16 / 32

`f4_f5_f6_cross_family_20261011__R13`	24 / 40
`f4_f5_f6_cross_family_20261011__R19`	8 / 24
`f6_ic_ecbench_ladder_20261005__n13`	24 / 72
`f6_ic_ecbench_ladder_20261005__n19`	24 / 72
`f6_ic_ecbench_ladder_20261005__n23`	not finished when the audit closed

Every failing row is an IC row (inherited-f4, f6-ic, and at K13 the matrix engines). All rho rows replay. In the root-finder session the four failing arms are ic-f4-m2, ic-f4-m3, ic-rootfind-m2 and ic-rootfind-m3, 24 rows each, so the root-finder arms fail too, which the published receipt says they did not. The committed ladder and cross-family audits passed on their own binaries, so the drift came with later code or a different host class; I did not bisect it. The same current binary adds a `row_ops` counter to `ic price`, which is the one counter difference I saw in the Table A and C replays and which changes nothing archived.

Counters at fault: solver word XORs, reductions, geometric additions and the phase GAE derived from them (the published seq 58 shows 301,961 word XORs now against 300,531 archived).

Rows they feed:

board row	consumes	status after the fresh replay
F6-IC E_0 ladder, m = 13, 19, 23 (F4 and F6-IC S/rho, median log2 W4/W6)	inherited-F4 and F6-IC solver counters directly	m = 13, 19: replay fails on every IC row. m = 23: replay-all had not finished; not covered
F4/F5/F6 cross-family (inherited F4 S, F6-IC S, W F5/F4, W F4 matrix/inherited, median log2 W inh/W F6)	the same counters	replay fails on every inherited-F4 and F6-IC row (K13 also matrix F4 and F5): 16/48, 16/32, 24/40, 8/24 identical
Latest frontier audit, replay block	reports the failure	says 48 F4 rows; fresh replay shows the wider failure
n9 F6-IC control	its own `phase_closure` replay certificate, not the ecbench counters	not affected by this failure; not replayed here
Tables A, B, C	`ic price` and `ic boundary` oracles (pair table, MITM, semaev), not F4	not affected: 18 of 18, 13 of 13 instances and 27 of 27 replays match
n37, n41/n53	counted MITM-Frobenius PDP, not F4	not affected; committed Linux audits 320/320 (cited, not re-run)

The effect on S: the root-finder study's F4 stage-count comparison is not current-binary evidence; for the ladder and cross-family rows, the S lower bounds include charged geometric additions, which are among the drifting counters. The recovered scalars verify in every case.

Rho reference, row group by row group

table	reference named by the protocol	strongest available?	priced in the same unit?	kernel advantage on the IC side
A (rule v3)	one-target rho as built, same process	strongest implemented: about 100 ns/step against about 200 ns/step for the standing fixture (the protocol's own admissibility check). Not the strongest possible: rho later gained 20% in instructions (2026-10-14).	yes, wall-clock units, same process	yes: unit = 1024-lane batched add; rho 32 lanes, 2.2 to 2.3 times the canonical step. IC-side kernel work cut the unit 2.3 to 3.2 times against section 23
B	matched negation-map rho, A = 2	yes for prime and random binary (negation is the only automorphism)	no: IC natives priced by pinned calibration, rho canonicalisations uncharged (45 to 4,196 per run)	none found
C	batch rho on the same 32 targets, canonical step, from S20	yes, at the canonical step	not the same process or session as the IC figure	none; rho gets the canonical (batched) step

The task statement mentions a rule v7 with rho at 9.4 to 10.6 units a step against 4.2 to 4.9. No rule v7 exists in this tree: `research/ic_tool_program/rule/` holds v2 (declared, never run) and v3. The v3 figures here are 9.7 to 11.8 against 4.6 to 5.4, the same pattern.

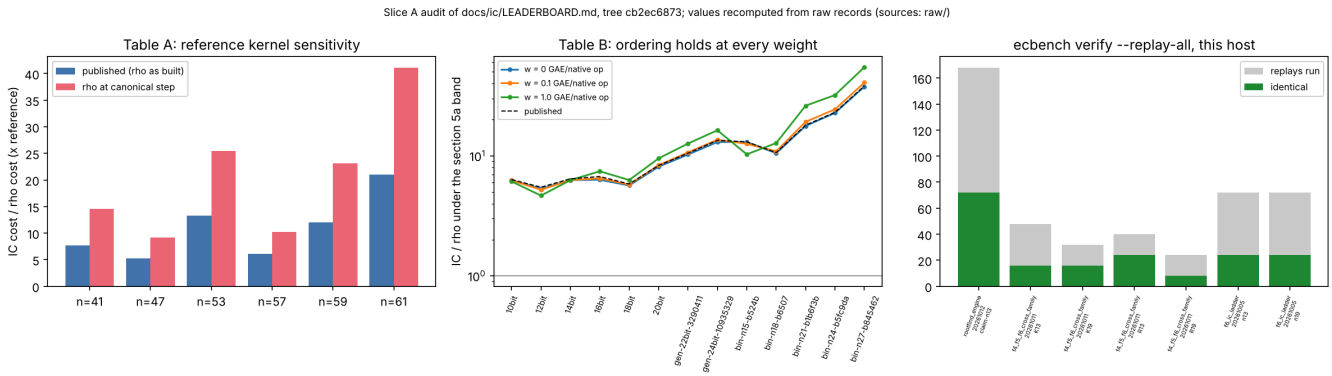


Figure 1: Audit summary. Left: Table A x reference as published and with rho at the canonical step. Middle: Table B x reference under the section 5a band at 0, 0.1 and 1 GAE per native operation, against the published value. Right: replays reproduced by ecbench verify replay-all on this host. Values from raw.

Pricing (section 5a)

- **A, C:** the unit is wall-clock for both arms, so there are no per-operation unpriced counters; the band was not run and is not directly applicable. `pricing_ok` is false because the board does not say that, and a counted check is missing.
- **B:** recomputed band (`raw/recompute_B_haiku.json`). IC / rho at w = 0, 0.1, 1: fp10 6.26, 6.24, 6.11 (published 6.34); fp24 13.0, 13.6, 16.3 (13.46); f2m27 37.7, 40.7, 54.8 (38.2). The order never changes; the magnitude can be 43 percent higher than published at w = 1.
- **n37:** `ecbench band` (`raw/band/`): every IC pair unrankable because field arithmetic, hashing and modular combination are named unpriced counters with no count. The K16-over-K42 ranking (0.349) is a ratio of lower bounds that section 5a does not allow to rank.
- **Ladder and cross-family:** F6-IC against F4 is "unresolved inside the band" at every size; rho is below every IC arm except F4 at m = 13, which is unresolved.
- **conductor-gap:** rho below IC at every weight.
- **n41/n53:** "incomplete" (IC arms did not verify).

Distance remaining (success criterion)

No row is below its matched rho. The distance is the x reference, recomputed:

table	x reference range	smallest	largest
A (rho as built)	5.23 to 20.97	f2m47	f2m61
A (rho at canonical step, recomputed)	9.13 to 41.13	f2m47	f2m61
B	5.47 to 38.21	fp12	f2m27
C	3.62 to 14.90	f2m41	f2m19

n37 K16 against strong rho: 5.03 counted (lower bounds), 4.70 complete-solve instructions; target-only online instructions 0.153 of rho. This audit claims no improvement.

Demotion flags (not applied)

1. **F6-IC ladder, m = 13, 19 (23 pending):** counters do not replay; freeze not demonstrable. Mark as historical, not current-binary evidence.
2. **F4/F5/F6 cross-family:** see table above (counters do not replay on current main); freeze not demonstrable.
3. **Frontier replay-block statement:** replace "48 inherited-F4 counter mismatches" with the fresh scope.
4. **n61 compact-orbit online bar:** keep as unvalidated until the 2026-10-12 sessions are linked from the board's sources.
5. **Table A header:** disclose the 32-lane versus 1024-lane kernel and show the canonical-step ratio beside it.
6. **Table C header:** it says rho is measured "in the same process"; for the isolated re-run it is S20's. Say so.
7. **n41 K16 256x quotient:** label as a survivor mean (23 of 80 verified).
8. **n37 K16 "lead":** label as unrankable under section 5a.

Per-row notes

See `verdicts.jsonl`; each record's `notes` field carries the row-level reasoning. Table A, B and C records are generated by `scripts/build_verdicts_ABC.sh` from `raw/`, the section records by `scripts/build_verdicts_sections.sh`.

Provenance and freeze order

First-commit order of protocol against raw data (`raw/provenance_haiku.jsonl`, identical to my own run):

evidence	protocol before raw data in git?	notes
Table A, rule v3	yes, 2026-10-06 vs 10-07, no later edits	
Table B, Round 5	yes, but BOUNDARY_TARGETS.md is edited 41 times afterwards	living document; recipe chosen as min of 9 on the measured targets
matched rho (B reference)	same commit	README says boundaries were committed first; not checkable
Table C (S22)	yes; v2 amendment before isolated runs	
n37 four studies	yes; session commit 5508af3 precedes data	k8-k16 analyzer frozen after data
conductor-gap, frontier rho, la_engines	yes (la_engines amended five times, adaptively)	
N83 holdout	yes: answers frozen 10:17 'before grid result', selection 11:55, results 14:21	
n9, F6 ladder, cross-family, n41/n53, frontier 20261013	same squashed commit	not demonstrable

Not covered, and the command that would cover it

- Ir rows (n37 K8/K16, online): sums of recorded per-workload values, not the raw callgrind zip. `ecbench callgrind-ir --prefix` on `RAW-CALLGRIND.zip` contents.
- Seeded replay of the 144-run cold bound and the n61 sessions (not located).
- R13 and R19 independent scalar checks: random binary curves carry no a, b in the record.
- Table C independent certificates: no target points archived.
- Native replay-all of the n37, n41/n53 sessions (Mac arm64 records): `ecbench verify --dir <session> --replay-all`.
- Bisecting the counter drift: replay the ladder session on the commit of each audit in `f6_ic_ecbench_ladder_20261005/audits/`.
- A repository-supported pricing band for Tables A and C: needs per-operation counts for the descent and PDP phases.

What this audit does not establish

It does not show any row is wrong in its arithmetic, and it does not show IC beats or approaches rho anywhere. It does not establish that the replay drift is a defect in the counters rather than an intended change that the old sessions predate. Verdicts say how far the evidence supports each row's published statement; they are not a research decision.